Session Objectives

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1. Be able describe the basic sacral landmarks and their locations
2. Be able to describe how to perform and interpret the different motion tests for sacral diagnosis.
3. Be able to describe the different sacral axes and their significances.
4. Be able to describe physiologic and non-physiologic sacral dysfunctions and what types of sacral dysfunction fall into each category and how they are named
5. Be able to apply the rules of L5 on the sacrum to make a sacral diagnosis if only given an L5 diagnosis.

Lumbosacral Junction

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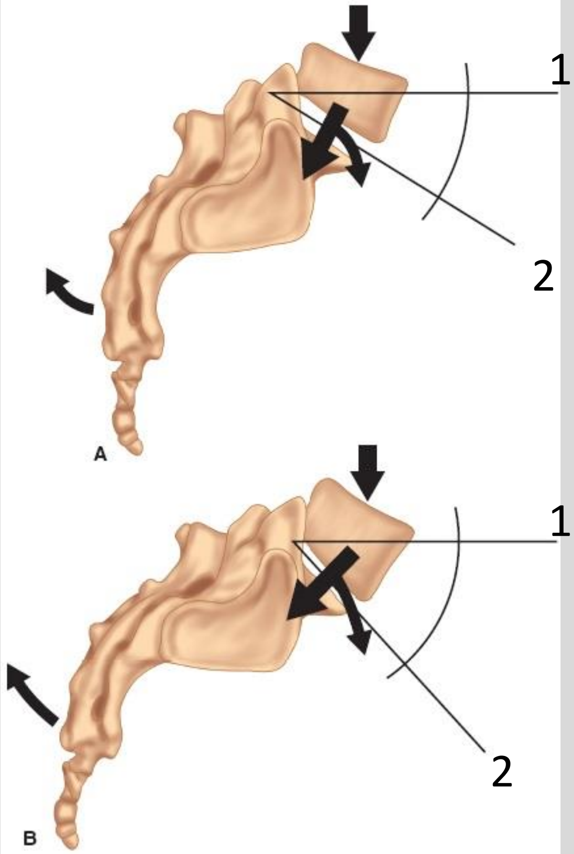

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Lumbosacral Angle (Ferguson's Angle) between:

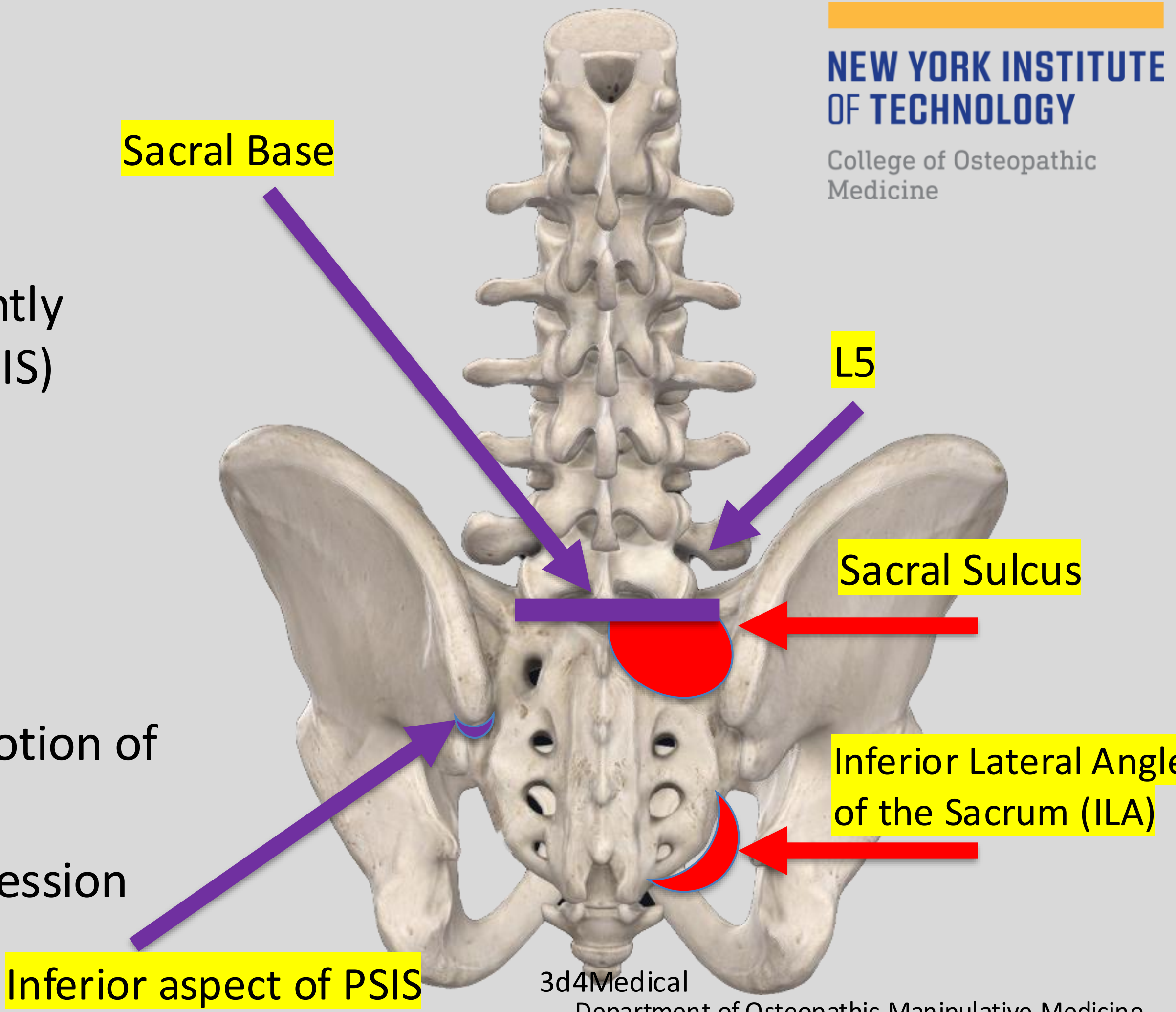
1-horizontal

2-sacral base

Normal= Approximately 25°-35°

An increase in the lumbosacral angle creates a greater anterior vector force, which increases the lumbosacral strain.

- These are the most important landmarks for making a sacral Diagnosis
  - Symmetry of:
    - Sacral Sulci (located slightly medil and superior to PSIS)
    - ILAs
  - Diagnosis of L5
  - Motion at:
    - Lumbosacral junction
    - SI joint with cephalad motion of ILAs
    - SI Joint with ASIS Compression and Seated Flexion test



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Axes of Motion of the Sacrum

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1 - vertical axis

2 - right oblique axis (gait, sacral torsions and sacral rotation)

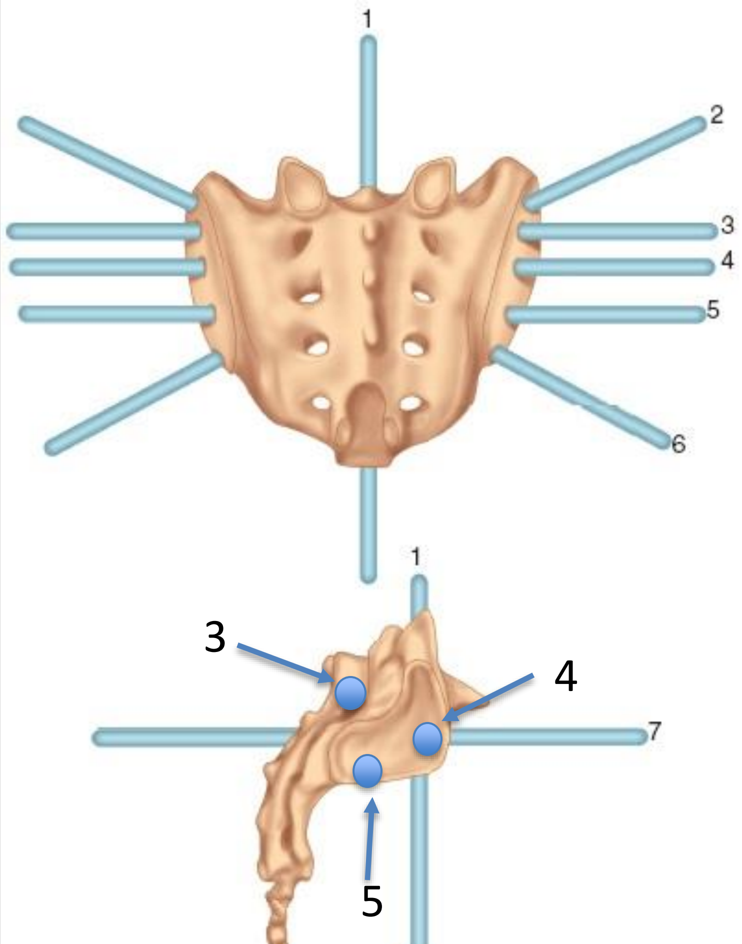
3 - Superior Transverse axis (respiratory and cranial-sacral motion)

*4 – Middle Transverse/*sacroiliac axis (Postural Motion)

5 - Inferior Transverse /iliosacral axis (Gait, Innominate rotation)

6 - left oblique axis (sacral torsions and sacral rotation)

7 - anteroposterior axis



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Transverse Sacral Axes

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1. **Superior Transverse Axis:** Respiratory and Cranio-sacral motion move around this axis

   In Inhalation:

   - Sacral base moves posteriorly (& superiorly)
   - Thoracic & pelvic diaphragms flatten, diameters increase
   - Spinal curves flatten

   In Exhalation:

   - Sacral base moves anteriorly (& inferiorly)
   - Thoracic & pelvic diaphragms dome, diameters decrease
   - Spinal curves become exaggerated

2. Middle Transverse Axis
3. Inferior Transverse Axis

Transverse Sacral Axes (Cont’)

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1. Superior Transverse Axis
   - Respiratory and Cranio-sacral Motion

2. Middle Transverse Axis
   - Postural Motion (seated or standing)
   - Sacral base
     - Anterior (nutation)
     - Posterior (counternutation)
   - Forward Bending and Backward bending around this axis

3. Inferior Transverse Axis

Nutation (nodding) vs. Counternutation

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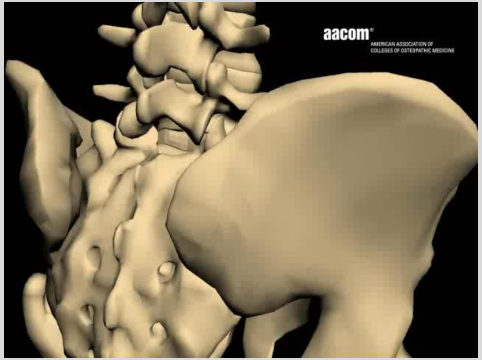
Axis: Middle Transverse

Other ways to say these:

- Anterior Nutation vs. Posterior Nutation
- (Sacral) Base Anterior vs. Base posterior
- Bilateral sacral flexion vs. B/L sacral extension



Bilateral Movement around a Transverse Axis

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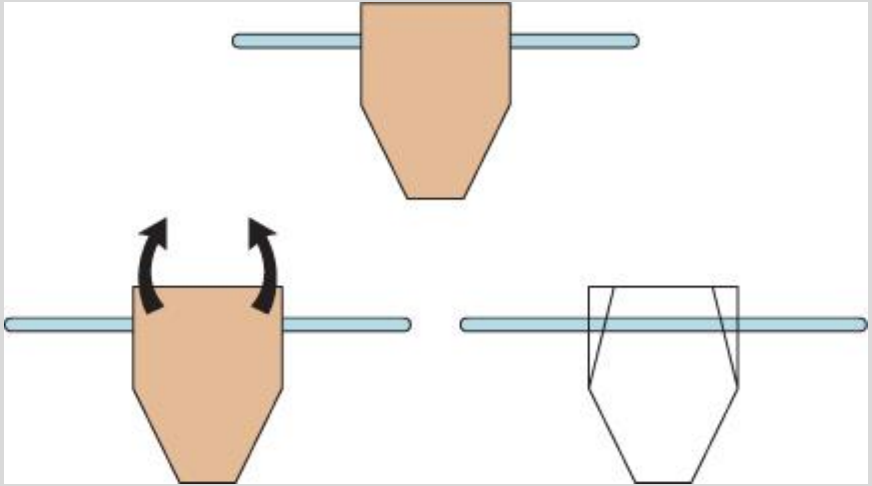
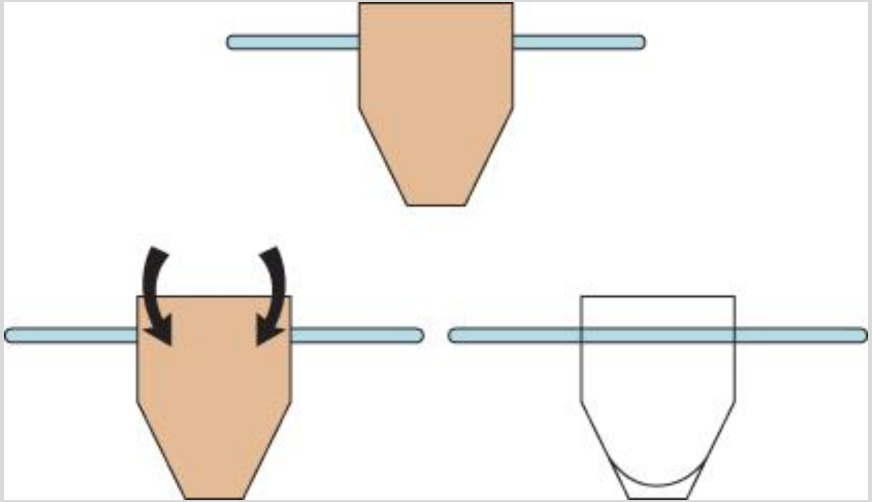


Sacral base is bilaterally flexing over a transverse axis



Sacral base is bilaterally extending over a transverse axis

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Unilateral Movement seems to move around a “Transverse Axis?”

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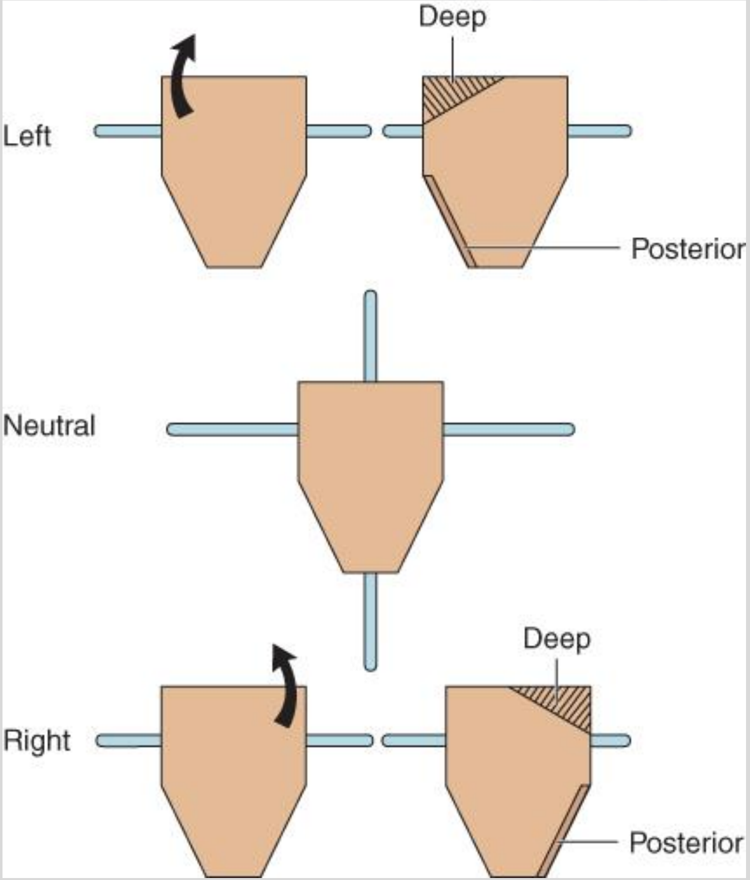
The sacrum can flex or extend unilaterally. The axis for this motion are not well understood. It is not really around a transverse axis. From a palalpatory standpoint, the following occurs. In unilateral flexion, the sacral base on the side of flexion will move anterior and the ILA on the same side will move posterior (and a little inferior). The opposite occurs in unilateral extension



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Oblique Sacral Axes

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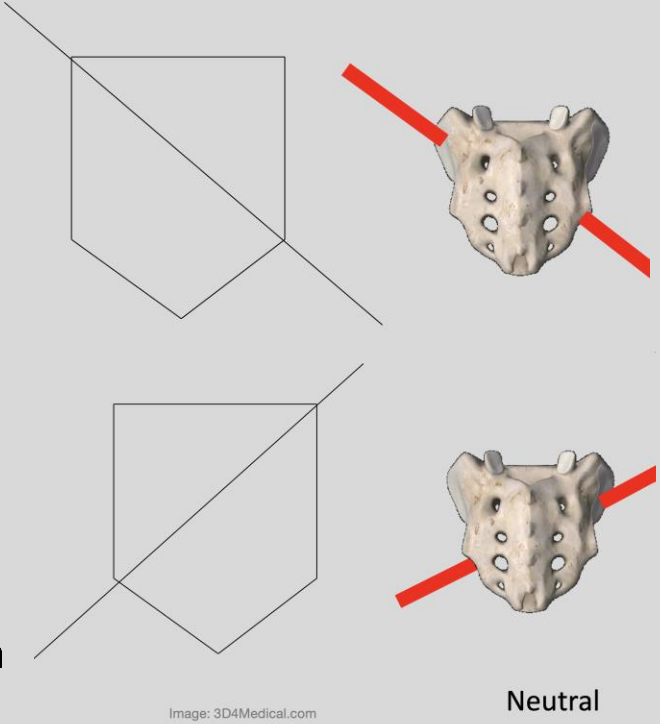
Left Oblique Axis

- Starts from the left superior sacroiliac articulation
- Diagonal to right inferior SI articulation

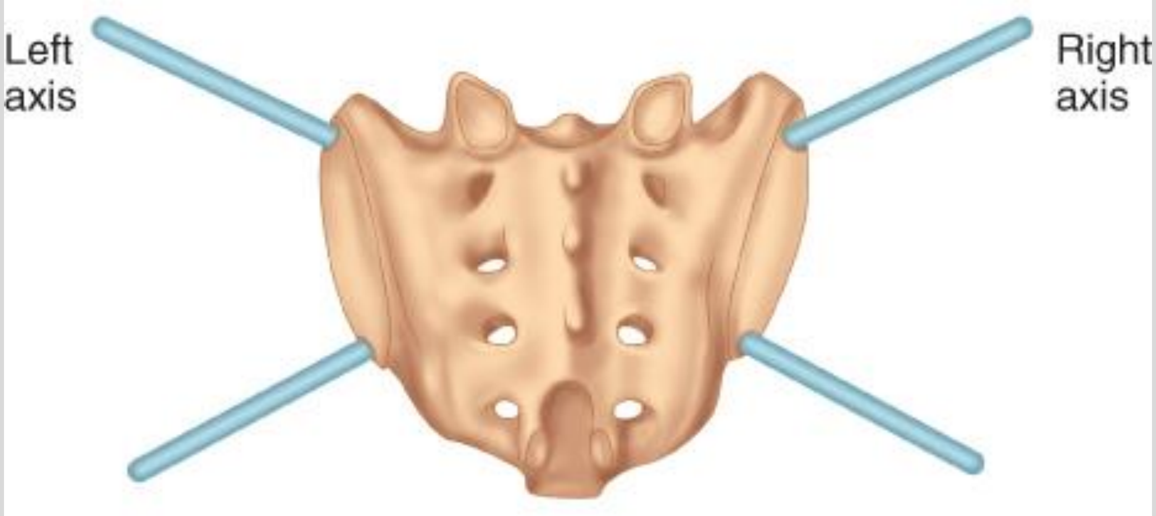
Right Oblique Axis

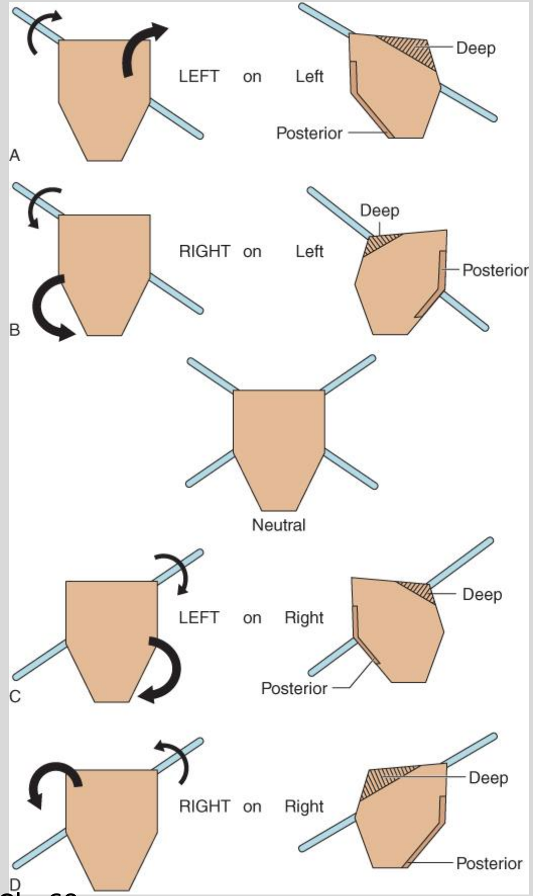
- Starts from the right sacral sulcus
- Diagonal to left infero-lateral angle of sacrum



Movement around an oblique axis

Examples are left rotation around a left oblique axis and left rotation around a right oblique axis





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Example: Left Sacral Rotation over oblique axes

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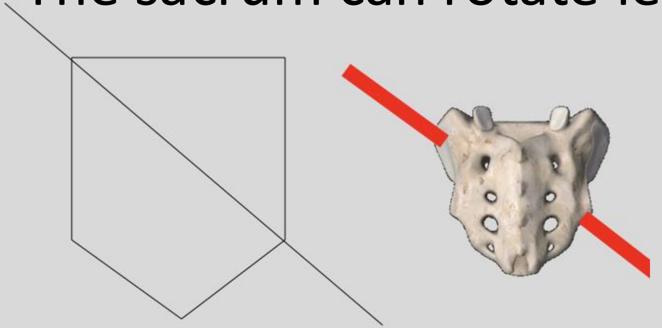
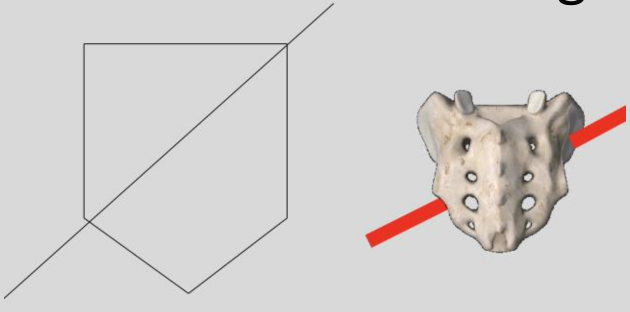
- The sacrum can rotate left on a left axis and left on a right axis





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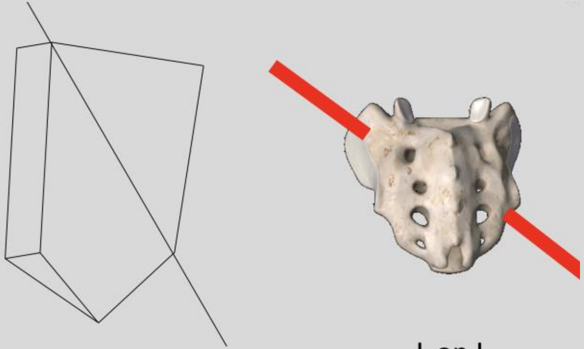
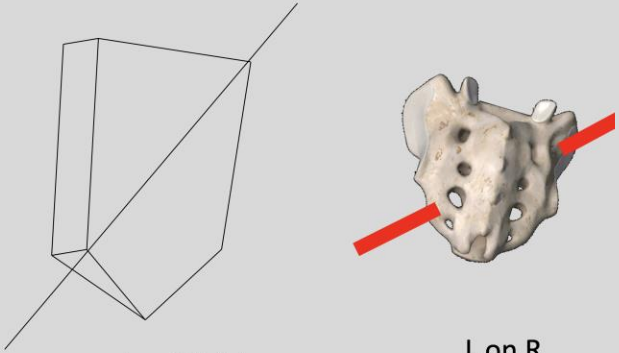
Neutral



L on L



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L on R

Physical Exam: Prone Static findings

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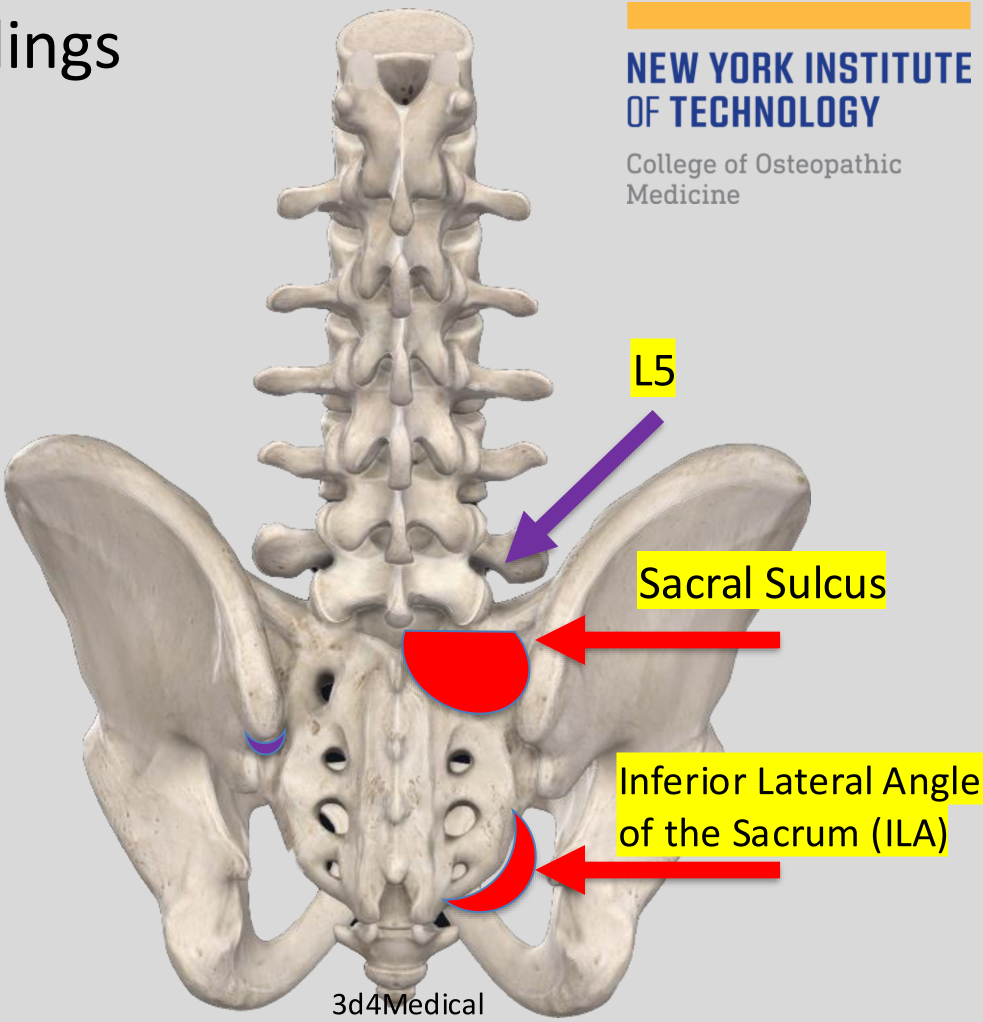
1. L5
2. Sacral Sulci
3. Inferior Lateral Angles (ILAs)



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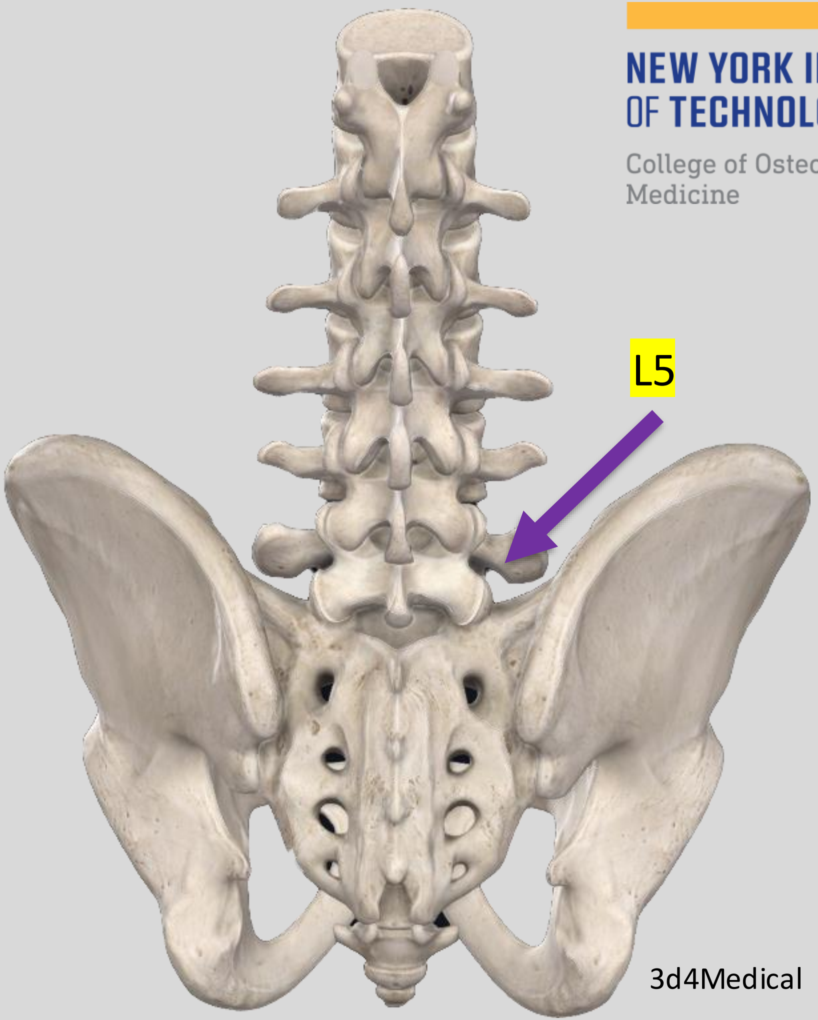
- Diagnose L5
  - Direction of L5 rotation influences certain sacral diagnoses:
  - Sacral Torsion Somatic Dysfunctions:
    - ❖ L5 & Sacrum rotate opposite
  - Sacral Rotation Somatic Dysfunctions:
    - ❖ L5 & Sacrum rotate same direction



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Check the symmetry of the Sacral Sulci

- Note which side is deeper (sacral base on that side is more anterior) & note which side is more shallow (sacral base on that side is more posterior)

- *Patient position:* prone.
- *Physician position:* varies according to the region examined.
- *Sacral sulci*
  - The physician stands at the side of the table and faces toward the patient's head.
  - The physician places her thumbs on each posterior superior iliac spine (PSIS).
  - The physician hooks her thumbs medially along the PSIS and brings her finger pads into the sacral sulci bilaterally
- The physician evaluates the relative depth of the sulci by two means: By palpating the depth of each sulcus by thumb position.
- By lowering her eyes to the level of the sulci for visual evaluation of depth.

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Check symmetry of ILAs

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- Note which side is posterior and inferior.
  - Remember if one side is posterior and inferior, the other side is anterior and superior.

1. Patient Prone
2. Physician's thumbs on inferior aspect of ILAs.
3. Compare which side is inferior
4. Then pace thumbs on posterior aspect of ILAs
5. Compare which side is more posterior

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Lumbosacral Spring Test

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- **Must be on L-S (lumbo-sacral) JUNCTION**
- **Be mindful of patient**
- **Gentle anterior force**

NORMAL SPRING TEST: If there is spring-ability at the L-S junction this is physiologic= a "*negative*" test:

- Forward moving sacral base (such as seen with forward torsions/rotations, and B/L sacral flexions)

ABNORMAL SPRING TEST = A positive (+) Test = ABSENCE of spring-ability :"steel-like resistance" – non-physiologic. Often result of sacral base being extended or backward

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Sphinx test (backward bending test)

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1. Interchangeable with Spring Test

Step 1: Patient prone – make note of static findings (sulci and ILAs)

Step 2: Physician places fingers at sulci and thumbs at ILA's B/L

Step 3: The patient is then instructed to either raise her body onto her elbows or if necessary onto palms to induce extension at the L-S junction. There should be an increased lumbar lordosis and sacral base should go anterior B/L during back extension.

(+) Test = Asymmetrical positioning of ILAs is maintained or worsened. This indicates a NON-physiologic sacral Somatic Dysfunction

(-) Test = ILA's become more symmetric when the patient's back is in extension.

Sphinx test (backward bending test)

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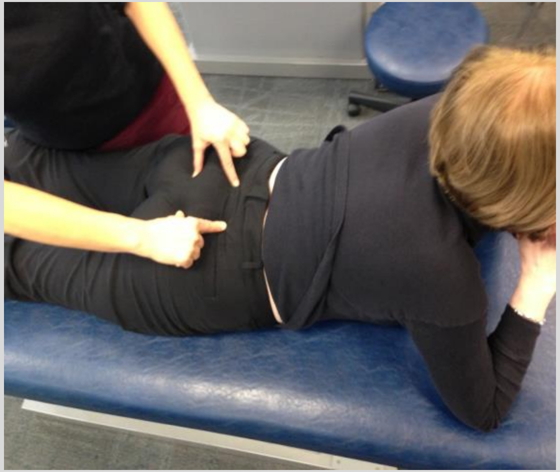

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Seated Flexion Test

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1. Patient sits with **feet flat on the floor**, knees parted
2. Physician is eye level with patient's PSIS's
3. Physician's thumbs are placed gently on inferior aspect of PSIS bilaterally
4. Patient bends forward from the waist with their arms dropping between their legs (not resting on their thighs)

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(+) Test = On the side that your thumb moves FIRST & FURTHEST Indicates <u>Sacral ON</u> Innominate somatic dysfunction (SD)

- **Side of positive test is used only in naming of (ipsilateral) unilateral sacral category of SD**
- **Side opposite of positive test is used for naming the axis for sacral torsions and rotations**
- But, the patients don't always follow the rules…will explain in lecture

(-) Test = Either: a) No SD

OR

b) Bilat. SD (B/L flexion or B/L extension SD)

ASIS compression test

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1. Patient is supine
2. Physician places her thenar/ hypothenar eminences along ASIS's B/L
3. Physician alternately introduces gentle but firm downward (A-P) pressure toward one S-I joint at a time from its ipsilateral ASIS contact. Physician allows the ilia to recoil anteriorly against gentle pressure
4. Determine ease of motion and compare bilaterally

(+) Test = Side that offers greater resistance to compression (left or right)

*ASIS Compression test is **NOT** specific for ilio-sacral vs. sacro-iliac dysfunction.

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ILA Motion Testing

(Another Test of Sacroiliac Joint (SIJ) Somatic Dysfunction Laterality)

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- Place thenar eminences on the inferior part of ILA's and fingertips at sulci
- Direct force cephalad from ILA into ipsilateral SIJ
- (+) test indicating dysfunction: a decrease in joint play on one or both sides and restriction of sacral motion.
- NOT used for naming sacral dysfunctions but a method of seeing which side may be restricted

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Physiologic vs. non-physiologic Categories of Sacral Sdysf(x)

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- **<u>Physiologic Dysfunction</u>**- somatic dysfunction produces a restriction within the normal range of motion of a joint

Examples:

- Forward Sacral torsions
- Forward Sacral rotations
- Bilateral Sacral Flexions

- **<u>Non-physiologic dysfunction</u>**-restriction of a joint in a position not obtained by usual physiologic motion. Usually traumatic but may occur in a patient with hypermobility

Examples:

- Backward sacral torsions
- Backward sacral rotations
- Unilateral Shears (sacral flexions/extensions)
- Bilateral Sacral Extensions

Non-physiologic dysfunctions should be treated prior to attempting OMT to other diagnoses

Physical exam → diagnosis

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1. Seated flexion test (Right or Left or neither?)
2. Sphinx test or Spring test (positive or negative?)
3. L5 Diagnosis (Type 1 or 2? SB? Rotation?)
4. Deep Sacral Sulci (R, L, both, neither?)
5. Posterior/Inferior ILA (R, L, both, neither?)

You will learn how to putthese findings into specific Diagnoses in the lab lecture and lab session.

Main Sacral SD categories w/ the possible outcomes:

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1) Bilateral Dysfunctions
- Bilateral flexion (physiologic)
- Bilateral extension (non-physiologic)

2) Unilateral Dysfunctions (both non-physiologic)
- Unilateral Flexion (aka sacral shear)
- Unilateral Extension

3) Sacral Torsions (L5 rotates opposite direction to the sacrum). Sacral torsions are named for **"direction of rotation of sacrum ON direction of the oblique axis"**
- Sacral base is FORWARD (physiologic)
  - L on L (= Left Rotation on a Left Oblique Axis)
  - R on R (= Right Rotation on a right Oblique Axis)
- Sacral base is BACKWARD (non-physiologic)
  - L on R (= Left Rotation on a Right Oblique Axis)
  - R on L (= Right Rotation on a Left Oblique Axis)

4) Sacral Rotations (L5 & sacrum rotate same direction). Similar to sacral tosrions, sacral rotations are SD named for **"direction of rotation of sacrum ON direction of the oblique axis"**

(same divisions as #3 above)

Pause... We need to mention:

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- Sacral Rotations and Torsions have an OBLIQUE AXIS
- UNILATERAL Sacral Dysfunctions – NO AXIS
- BILATERAL Sacral Dysfunctions have a TRANSVERSE (middle) AXIS

Sacral Torsions and Rotations move about an OBLIQUE AXIS

UNILATERAL Sacral Dysfunctions have NO Real AXIS to speak of

BILATERAL Sacral Dysfunctions move about a middle TRANSVERSE AXIS

NEED TO KNOW

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- Sacral Rotations and Torsions have Deep Sulcus and Post-inf ILA on OPPOSITE sides

- UNILATERAL Sacral Dysfunctions have Deep Sulcus and Post-Inf ILA on SAME side

- BILATERAL Sacral Dysfunctions are either flexed (nutated) or extended (counter-nutated)

DIAGNOSING SACRAL TORSIONS BASED ON L5 (You will need to memorized this slide and the next 3 slides. This is a different way of diagnosing sacral torsions which you may see on future exams. In the Lab, we will be going over how to diagnose sacral dysfunctions using the sacral landmarks and and other findings as discussed earlier in this presentation.)

- **L5** will be **rotated opposite** the direction of the **sacrum (if not, L5 must be treated first)**
- **Seated flexion test is (+) opposite the axis of rotation** in a torsion**.**
- **L5** will be **sidebending** to the same side as the oblique **Axis**

DIAGNOSING SACRAL TORSIONS BASED ON L5 (continued)

- **Forward** torsions are associated with **neutral lumbar** mechanics of **L5** (ie. L5 N S_LR_R with a L on L FST and L5 N S_RR_L with a R on R FST)

- **Backward** torsions are associated with **non-neutral lumbar** mechanics of **L5** (i.e L5 ERS_R or FRS_R with a L on R BST, and L5 ERS_L or FRS_L with a R on L BST)

- No sacral rotations using this model since the sacrum is rotating opposite L5

DIAGNOSING SACRAL TORSIONS BASED ON L5 (continued)

What does this mean?

- If you are given that L5 ERS right, this means that L5 has a non-neutral (type 2) dysfunction therefore:
  - The Axis is on the right (the side of the SB)
  - The + seated flexion test is on the left (opposite the axis)
  - Sacrum is rotating left (rotates opposite L5)
  - Diagnosis is Left on Right Sacrum (rotation Left on a Right axis)
  - Left on Right Backward Sacral Torsion (if Rotation and axis are opposite, it is a backward torsion
  - The same for L5 FRS right

DIAGNOSING SACRAL TORSIONS BASED ON L5 (continued)

What does this mean? (continued)

- Given that L1-L5 N S left, R right, this means that L5 has a neutral (type 1) dysfunction therefore:
  - The Axis is on the left (the side of the SB)
  - The + seated flexion test is on the right (opposite the axis)
  - Sacrum is rotating left (rotates opposite L5)
  - Diagnosis is Left on Left Sacrum (rotation Left on a Left axis)
  - Left on Left Forward Sacral Torsion (if Rotation and axis are the same direction, it is a Forward torsion)

Session Objectives

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1. Be able describe the basic sacral landmarks and their locations
2. Be able to describe how to perform and interpret the different motion tests for sacral diagnosis.
3. Be able to describe the different sacral axes and their significances.
4. Be able to describe physiologic and non-physiologic sacral dysfunctions and what types of sacral dysfunction fall into each category and how they are named
5. Be able to apply the rules of L5 on the sacrum to make a sacral diagnosis if only given an L5 diagnosis.

Session Objectives

At the conclusion of the session, the students should be able to:

1. Understand the movement of the sacrum on the different sacral axes of rotation and the significance of each axis.
2. Understand L5 rules as they relate to sacral torsions
3. Understand all sacral special tests for diagnosis
4. Be able to put information on axes and special tests together to make a sacral diagnosis

Transverse Axes

- Superior
  - Through S2 attachment of dura
  - Primary Respiratory Motion
  - Gross respiratory motion
- Middle
  - Through S2 vertebral body (Also, center of mass of human body)
  - Sacral motion on pelvis
  - Axis of bilateral sacral flexion or extension dysfunctions
- Inferior
  - Through S3
  - Innominate movement relative to sacrum
  - Axis of innominate dysfunctions

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Oblique Axis

- From superior SIJ on one side to inferior SIJ on opposite side
- Sacral torsions and rotations
- Named for movement ON axis
- Reference of movement is the anterior sacral body
- Can be torsioned
  - Left on Left (anterior body MUST be going FORWARD)
  - Right on Left (anterior body MUST be going BACKWARD)
  - Imagine a pair of eyes on anterior sacrum… looking left or looking right)
- Can be rotated
  - Same rotation as side of axis is forward, and opposite is backward (don't need to memorize this!)

Torsions vs. Rotations

- Torsion = Twisting
  - Wringing a towel… twisting means turning ends in opposite directions
  - Sacrum turns in opposite direction of L5

- L5 Rules for sacral torsion:
  - Based on assumption that sacrum is in a torsion…

- Rotation L5 IS side OPPOSITE of sacral ‘rotation’ (torsion)- by definition
- **Side bending IS side of oblique axis**
- L5 Type I dysfunction = Forward/Type II dysfunction = Backward

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Physiologic vs. Nonphysiologic

- Normal motion
  - Sacrum moves forward and backward (nutation and counternutation) as well as obliquely during normal everyday movements

- Abnormal motion

  Physiologic Dysfunctions:
  - DYSFUNCTIONAL sacral movement is ’stuck’ in a forward direction
  - Bilateral sacral flexion dysfunction, forward sacral torsion or rotation

  Non-physiologic Dysfunctions:
  - DYSFUNCTIONAL sacral movement is ‘stuck’ in a backward direction
  - Bilateral sacral extension dysfunction, backward sacral torsion or rotation

  - Sacral shears (unilateral)… forward OR backward

Special Tests

- Spring Test
  - Patient prone
  - Pushing on lumbosacral junction
  - Normal response is a feeling of ‘springing’ down (forward) and back up
  - If it doesn’t spring down/forward (anterior)… it’s stuck backward (posterior)
- Sphinx Test
  - Patient prone
  - Palpate at sacral sulci and ILA
  - Patient goes onto elbows or hands → increasing lordosis and lumbosacral angle = sacral flexion
  - If dysfunction is forward, sacral flexion is going into freedom and it evens out
  - If dysfunction is backward, sacral flexion is the barrier and it gets worse

Special Tests

- Seated Flexion test
  - Thumbs on PSIS, on skin
  - Patient seated with feet flat on floor and flexes trunk forward
  - Positive on side of PSIS moving first and farthest
  - This asymmetry indicates some dysfunctional movement on that side
  - Movement happens ON an axis (around it) so axis is on opposite side of positive finding

  - Dysfunctional movement…

Application

- Seated flexion test: Positive on right / Tells you…
  - Right shear or a some torsion/rotation ON a left axis)
- Spring test: Negative / Tells you…
  - (Forward unilateral flexion or forward torsion/rotation)
- Sacral sulci: Deep on right
- ILA: posterior on left
- L5 NSlRr
- Left (sacrum) ON left (axis) forward sacral torsion

Application

- Seated flexion test: Positive on left / Tells you…
  - Left shear or torsion/rotation ON a right axis)
- Sphinx test: Positive / Tells you…
  - Backward shear or torsion/rotation
- Sacral sulci: Deep on right**
- ILA: Posterior on left**
- Left (sacrum) ON right (axis) backward sacral torsion
- All motion is relative…

Sulci and ILAs

- Motion is relative
  - A ‘deep’ sulcus is only deep relative to axis bearing sulcus, which feels comparatively shallow
  - A ‘posterior’ ILA feels posterior only because you’re comparing it to the ILA on the other side, which itself feels ‘anterior’
  - Your fingers only feel a difference… which finger is right?

  - Know whether its forward or backward (Spring or Sphinx)
  - Know where axis is (Seated flexion test, L5 position)
  - Understand deep vs. shallow relativity
  - Apply palpatory findings

Sulci and ILAs

- Motion is relative
  - A ‘deep’ sulcus is only deep relative to axis bearing sulcus, which feels comparatively shallow
  - A ‘posterior’ ILA feels posterior only because you’re comparing it to the ILA on the other side, which itself feels ‘anterior’
  - Your fingers only feel a difference… which finger is right?

  - Know where axis is (Seated flexion test or L5 position)
  - Know whether its forward or backward (Spring or Sphinx) Understand deep vs. shallow relativity
  - Apply palpatory findings